



DEPARTMENT OF COMPUTER SYSTEMS ENGINEERING
MEHRAN UNIVERSITY OF ENGINEERING & TECHNOLOGY, JAMSHORO
Microprocessor and Interfacing
Project Report

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Creating a retro-Nintendo Emulation System using ESP-32 WROOM DA Module

Introduction:

The project is aimed to create a Nintendo emulator using the ESP32 microcontroller, allowing users to play classic Nintendo games on a breadboard setup. The ESP32 is a powerful microcontroller that combines Wi-Fi and Bluetooth capabilities, making it suitable for various applications, including emulation. The ESP32 was chosen for its processing power, connectivity options, and GPIO pins suitable for interfacing with input devices and display components.

Objectives:

- Create a functional Nintendo emulator on the ESP32 platform.
- Assemble a breadboard rendition of the Nintendo console using economically viable components.
- Present a thorough procedural guide for replication purposes.

The ESP32:

The ESP32 WROOM module, based on the ESP32 system-on-chip (SoC) developed by Espressif Systems, is a highly versatile microcontroller with integrated Wi-Fi and Bluetooth capabilities. Its architecture consists of a dual-core Xtensa LX6 microprocessor, clocked up to 240MHz, with multiple peripheral interfaces including SPI, I2C, UART, and GPIOs for flexible connectivity. It features built-in security features like secure boot and encrypted flash memory, making it suitable for various IoT applications. With its low power consumption and advanced features, the ESP32 enables efficient and cost-effective development of IoT devices, offering ample computational power and connectivity options in a compact package.



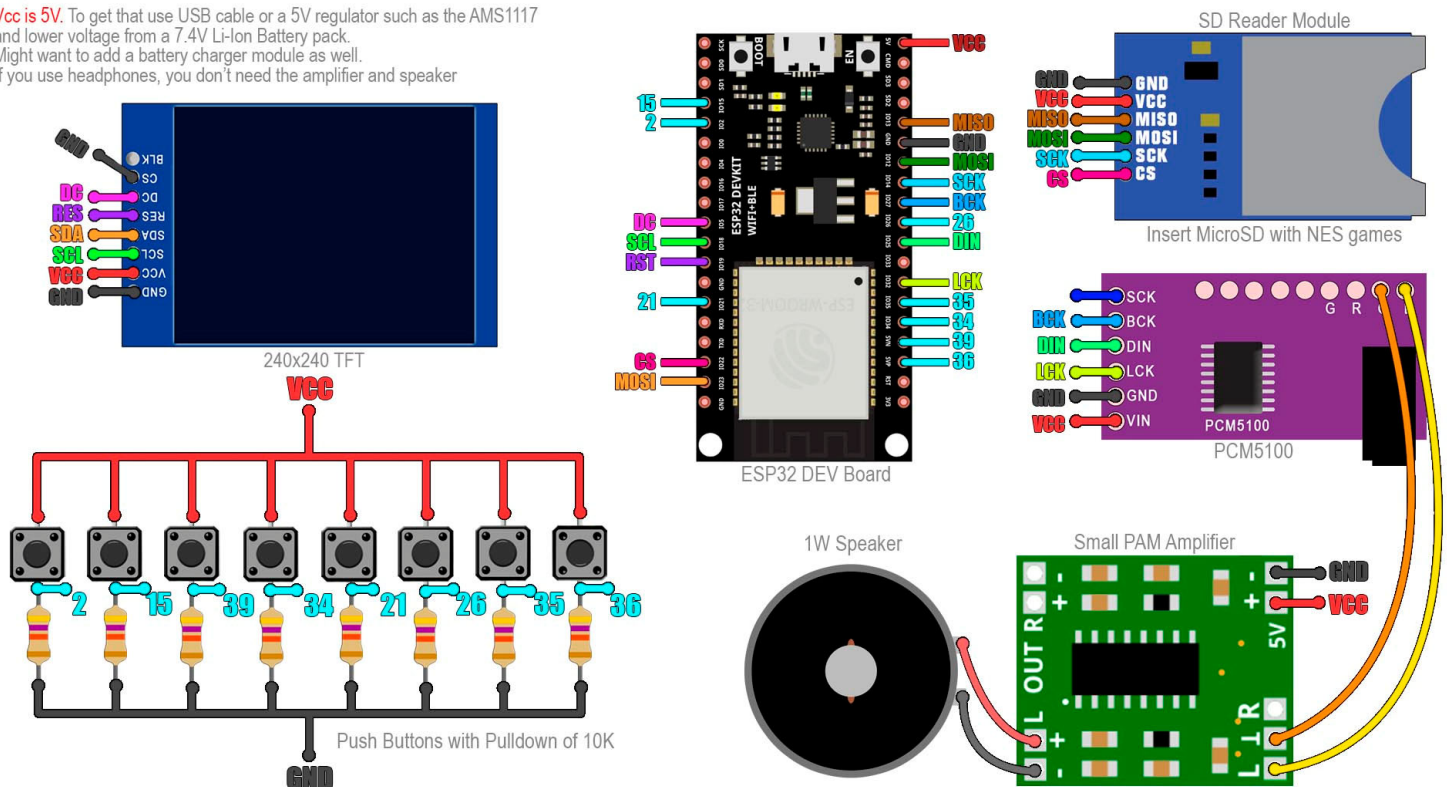
*Pins SCK/CLK, SDO/SD0, SDI/SD1, SHD/SD2, SWP/SD3 and SCS/CMD, namely, GPIO6 to GPIO11 are connected to the integrated SPI flash integrated on ESP-WROOM-32 and are not recommended for other uses.

Materials Used:

- ESP32 development board
- SD card module
- 240x240 IPS TFT display
- Various jumper wires
- Nintendo game ROMs
- Input devices for controller (push buttons, etc.)
- 7x 10K Ohm Resistors

Diagram Representation of Nintendo Emulator on ESP32

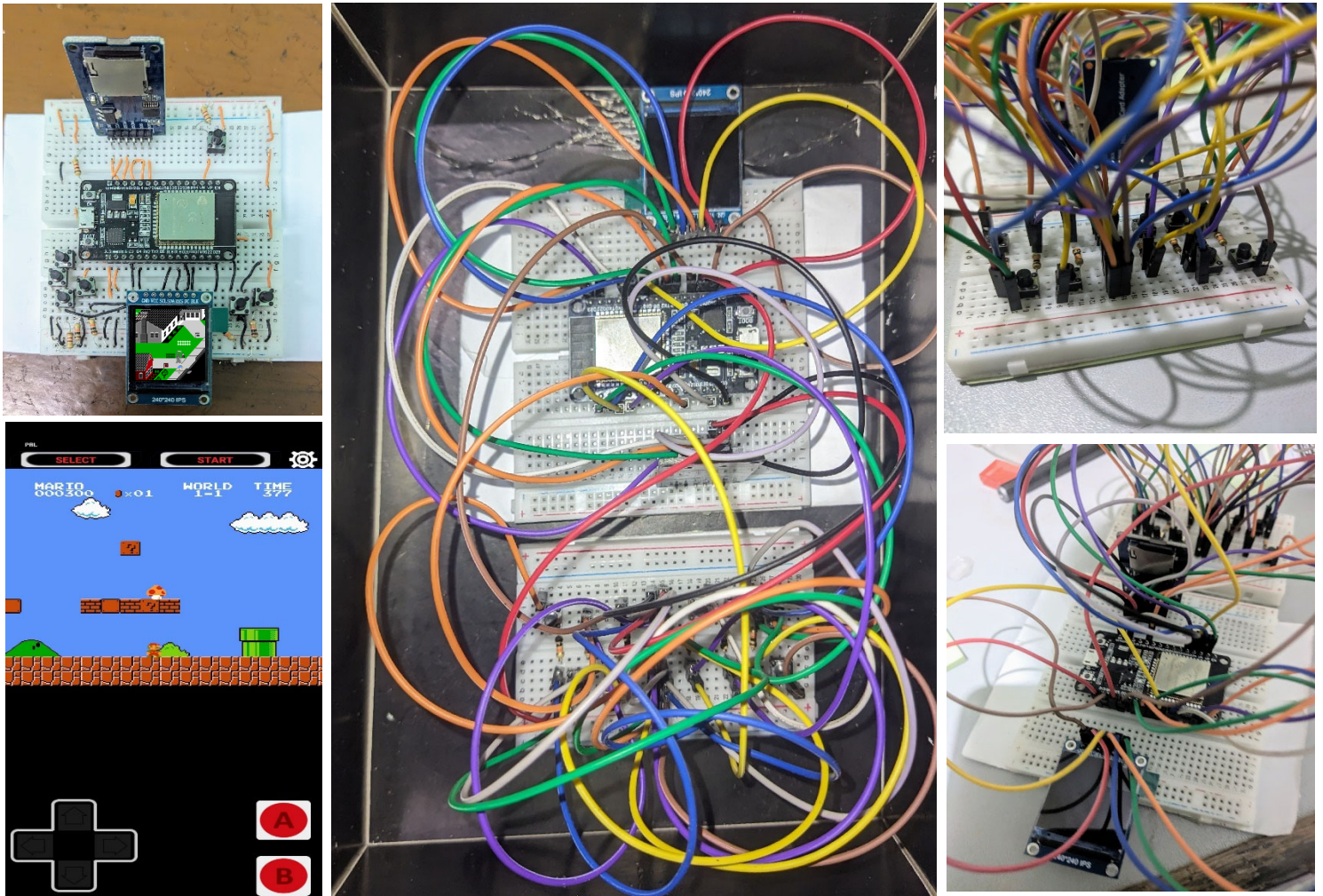
Vcc is 5V. To get that use USB cable or a 5V regulator such as the AMS1117 and lower voltage from a 7.4V Li-Ion Battery pack. Might want to add a battery charger module as well. If you use headphones, you don't need the amplifier and speaker



Methodology:

- Setting Up Components:** Install Arduino IDE and requisite ESP32 board support. Establish connections between the ESP32 development board, SD card module, and TFT display as per specifications. Download essential libraries for the project and integrate them into the Arduino IDE.
- Emulator Configuration:** Obtain the emulator code:
<https://www.dropbox.com/scl/fi/cuh2zgaw61sr7o7w1azgu/NES-ESP32-ino.txt?rlkey=ntvtwnp2wla0b0srepxhs4c67&st=lgvc7apo&dl=0>
Install mandated libraries and extract them into the Arduino libraries folder. Customize pin connections in the Arduino sketch as instructed. Upload the configured sketch to the ESP32 board.
- Loading Games:** Create an "NES" folder on the SD card and procure Nintendo game ROMs. Insert the SD card into the module and connect it to the ESP32. Access games through the emulator upon system activation.

4. **Results:** Successful creation of a Nintendo emulator on the ESP32 platform. Construction of a fully operational breadboard iteration of the Nintendo console using cost-effective components. Validation of emulator functionality through testing with various Nintendo games.



Future Enhancements and Applications of this Project:

Creating a retro-Nintendo emulation system using the ESP32 WROOM DA Module could lead to applications in nostalgic gaming experiences, education, hobbyist projects, museum exhibitions, and potentially as consumer products due to the growing interest in retro gaming.

- Integration of sound capabilities into the emulator.
- Refinement of the breadboard Nintendo design for enhanced aesthetics and durability.
- Exploration of additional features such as save states and multiplayer support.

Conclusion:

The project demonstrates the viability of crafting a Nintendo emulator on the ESP32 microcontroller and fabricating a breadboard version of the classic gaming console. The project outcomes highlight not only the technical capabilities of the ESP32 microcontroller but also the affordability and accessibility of the components utilized. Moreover, the comprehensive procedural guide ensures that individuals with varying levels of technical expertise can undertake similar projects with confidence. Overall, this project exemplifies innovation and invites enthusiasts to explore the realms of retro gaming and microcontroller development.